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RESEARCH ARTICLE

Low-Light Object Detection via Image-Level Illumination-Conditioned Modulation

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ABSTRACT Object detection in low-light scenes is challenged by weak contrast, amplified noise, and spatially nonuniform exposure, which jointly degrade feature representation and increase missed detections. Existing enhancement-first pipelines can introduce additional computation or alter task-relevant appearance, while detectors with fixed feature transformations cannot explicitly adapt to the illumination state of each input. This article proposes Illumination-Conditioned Modulation YOLO (ICM-YOLO), a compact detector that incorporates illumination awareness at both the image and prediction levels. An Image-Level Illumination Descriptor provides a shared global condition, Conditioned Dynamic Modulation C3k2 adapts semantically enriched neck features, and a Classification-Oriented Luminance-Ratio Adaptation Head recalibrates category evidence under local exposure variation while preserving the localization pathway. Experiments on ExDark show that ICM-YOLO achieves 63.75% mAP@50, improving the YOLO26n baseline by 1.32 percentage points, with an approximately 2.0% increase in parameters. Ablation results support the effectiveness of classification-oriented adaptation and selective neck conditioning. On visible-only M3FD, ICM-YOLO yields a small gain over YOLO26n. These results indicate that selective illumination conditioning offers an effective and lightweight approach to low-light object detection.

INDEX TERMS Feature modulation, illumination adaptation, low-light object detection, luminance-aware detection, real-time object detection.

I. INTRODUCTION

Object detection is a core capability in surveillance, autonomous transportation, robotic perception, and mobile vision, yet these applications must frequently operate under street lamps, vehicle headlights, and weak ambient light. Low illumination causes more than a uniform decrease in brightness: short exposure and high sensor gain amplify noise, camera processing suppresses dark-region detail, local highlights coexist with severe underexposure, and longer exposure can introduce motion blur. The resulting loss of edges, texture, and color disproportionately affects small or occluded objects. NightOwls [1] and NOD [2] demonstrate these difficulties in nighttime pedestrian detection and extreme low-light scenes. Moreover, perceptual enhancement does not consistently improve recognition under poor visibility [3]. Reliable low-light detection must therefore address both degraded visual evidence and illumination-dependent feature variation.

Modern detection methods provide strong foundations for addressing these conditions. Faster R-CNN established the proposal-based two-stage paradigm [4], whereas RetinaNet demonstrated effective dense prediction through focal loss [5]. Feature pyramid networks made multi-scale representation a standard component for objects with large size variation [6]. More recent one-stage detectors emphasize efficient end-to-end inference, making them attractive for resource-constrained nighttime applications. However, models developed primarily on normally exposed imagery generally apply the same learned transformations to

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every input. When illumination changes substantially, fixed feature processing may amplify noise in shallow layers or discard faint object boundaries during downsampling, motivating detector-side mechanisms that adapt representation to the observed lighting condition.

The YOLO family provides an efficient basis for studying this problem. YOLOv8 offers an anchor-free real-time implementation [7], while YOLOv9 improves information preservation through programmable gradient information and GELAN [8]. YOLOv10 advances NMS-free end-to-end detection [9], and YOLO11 continues the efficiency-oriented Ultralytics line [10]. More recently, YOLOv12 introduces an attention-centric architecture [11], and the Ultralytics YOLO26 release simplifies deployment through DFL-free box regression and an optional NMS-free one-to-one head [12]. Despite their different designs, these detectors still apply fixed learned transformations after training. Under low light, however, shallow features become sensitive to noise and weak contrast, while downsampling can remove faint boundaries from deeper features. The detector consequently lacks an explicit mechanism for sharing the illumination state across its feature hierarchy.

Low-light image enhancement is a natural but indirect response. LIME estimates an illumination map through a model-driven prior [13], whereas Retinex-Net learns reflectance–illumination decomposition [14]. Zero-DCE avoids paired supervision by estimating image-specific enhancement curves [15], and Retinexformer combines illumination guidance with transformer-based restoration [16]. These methods improve visibility, but reconstruction or perceptual quality is not equivalent to detection accuracy. Enhancement can amplify noise or alter class-relevant appearance, and a separate restoration network adds computation before the detector.

Task-oriented low-light detectors integrate degradation handling more directly. MAET jointly learns degradation transformations and object representations [17]. Detector-side approaches instead adapt feature extraction and fusion: 3L-YOLO emphasizes lightweight multiscale context [18], LLD-YOLO dynamically combines shallow and deep features [19], and Dark-YOLO introduces multiple attention mechanisms for dark scenes [20]. In these detector-side methods, the control signals are derived primarily from the feature tensor currently being processed. A compact image-level description of illumination is not explicitly reused across semantic levels, and photometric adaptation is not consistently separated from localization.

A closely related line of work conditions detection on a global image representation. IA-YOLO and GDIP-YOLO use such a representation to parameterize differentiable image-processing operators [21], [22], whereas DENet and PE-YOLO enhance Laplacian-pyramid components before detection [23], [24]. Both groups therefore adapt the detector by transforming the input image before detection. FeatEnHancer instead refines and aggregates hierarchical features through scale-aware attention [25]. In contrast, ICM-YOLO leaves the input image unchanged, reuses a compact illumination descriptor at selected neck levels, and restricts spatially varying recalibration to the classification pathway.

Conditional feature modulation provides a complementary perspective. FiLM generates channel-wise affine parameters from an auxiliary conditioning representation [26], allowing shared features to respond to sample-specific context without changing their spatial resolution. Illumination is well suited to this formulation because its global component affects color, texture, contrast, and activation magnitudes across several semantic levels. However, effective use in a pretrained lightweight detector requires resolving three questions: how to summarize illumination compactly, where to apply modulation without erasing fine detail, and how to prevent photometric adaptation from unnecessarily disturbing localization-sensitive features.

This work addresses these questions using YOLO26n, the nano-scale model of the Ultralytics YOLO26 release [12]. The proposed Illumination-Conditioned Modulation YOLO (ICM-YOLO) combines a shared image-level illumination descriptor, selective P4/P5 feature modulation, and classification-specific adaptation while retaining the input image, baseline feature-pyramid topology, and localization stream. CDM adopts the residual FiLM affine form; the contribution lies in the conditioning signal, its selective placement, and the classification–regression separation rather than in a new modulation operator.

The main contributions of this article are summarized as follows.

1) **Image-Level Illumination Descriptor (IID):** A lightweight descriptor is introduced to represent the global illumination state of each image. By combining learned context with explicit luminance statistics, IID supplies a shared and interpretable conditioning signal to multiple detector stages.
2) **Conditioned Dynamic Modulation C3k2 (CDM-C3k2):** An illumination-conditioned neck module is developed to adapt semantically enriched features while retaining the structure and tensor interface of the baseline C3k2 block. The module enables sample-dependent feature transformation with limited additional complexity.
3) **Classification-Oriented Luminance-Ratio Adaptation Head (CLRA Head):** A task-specific detection head is designed to model spatially nonuniform exposure through activation-based luminance-proxy cues. It recalibrates classification evidence while preserving the original box-regression pathway, thereby separating photometric adaptation from localization.

II. METHODOLOGY

A. YOLO26n BASELINE ARCHITECTURE

YOLO26 is the detector family released by Ultralytics in January 2026 with version 8.4.0 of the `ultralytics` package [12], and YOLO26n is its nano-scale variant.

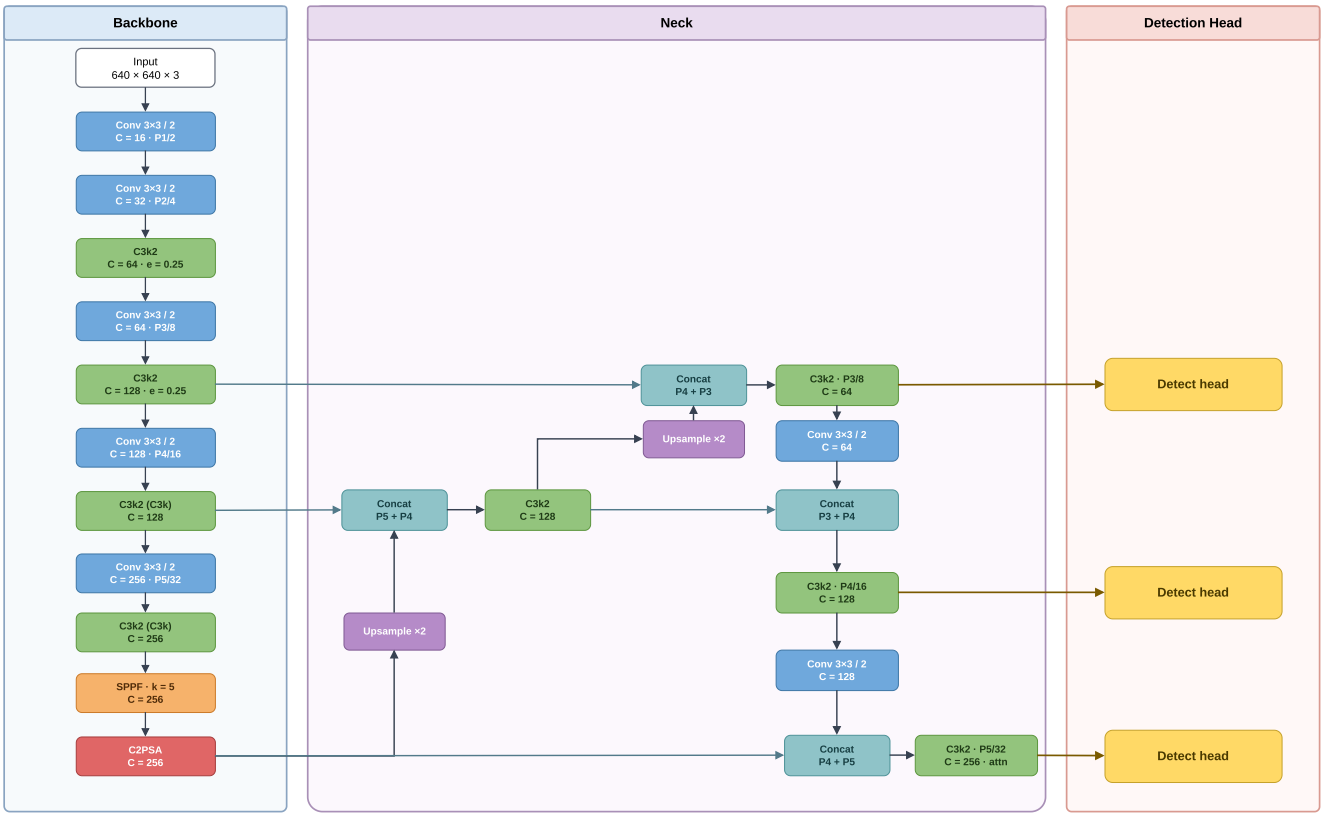


FIGURE 1. Architecture of YOLO26n.

The number 26 is a release designation within the Ultralytics line, which also includes YOLOv8 and YOLO11, rather than a successor numbering of independently developed detectors such as YOLOv9, YOLOv10, and YOLOv12. Throughout this article, YOLO26n denotes the unmodified official nano configuration initialized from the COCO-pretrained checkpoint `yolo26n.pt`. It is selected as the host baseline because its compact parameterization facilitates a controlled evaluation of illumination adaptation. Given an input image $\mathbf{I} \in \mathbb{R}^{3\times H\times W}$, the backbone constructs a hierarchical representation through alternating stride-2 convolutions and C3k2 blocks. The spatial resolution is progressively reduced from $H \times W$ to $H/32 \times W/32$, while the channel capacity increases from 16 to 256. C3k2 blocks refine the features at each stage through compact cross-stage feature reuse, balancing representation capacity and computational cost. The intermediate outputs at strides 8 and 16 retain comparatively detailed spatial information, whereas the stride-32 output encodes stronger high-level semantics.

At the deepest stage, spatial pyramid pooling-fast (SPPF) aggregates context over multiple effective receptive fields without further reducing the feature-map resolution. The subsequent C2PSA block recalibrates the high-level representation before multiscale fusion. As shown in Figure 1, the neck adopts bidirectional FPN/PAN-style aggregation. In the top-down path, the stride-32 feature is upsampled and concatenated with the corresponding stride-16 backbone feature. After C3k2 refinement, the resulting feature is upsampled again and fused with the stride-8 backbone feature. This process transfers semantically rich deep information toward the higher-resolution levels, which is important for recognizing small or weakly exposed objects.

The bottom-up path strengthens localization information in the opposite direction. The fused stride-8 feature is downsampled and concatenated with the intermediate top-down stride-16 feature to form the final P4 representation. P4 is then downsampled and aggregated with the deep feature to obtain P5. The neck therefore outputs three feature maps, $\{\mathbf{P}_3, \mathbf{P}_4, \mathbf{P}_5\}$, at strides 8, 16, and 32, respectively. P3 emphasizes spatial detail, P5 provides the broadest contextual support, and P4 offers an intermediate balance between localization and semantics.

The detection head receives all three scales and produces category and bounding-box predictions through separate classification and regression pathways. Multiscale prediction allows the detector to accommodate substantial variations in object size while preserving an efficient feed-forward structure. In this work, the backbone stages, FPN/PAN aggregation routes, and three-scale prediction topology of YOLO26n are retained. Keeping these components unchanged provides a controlled architectural reference

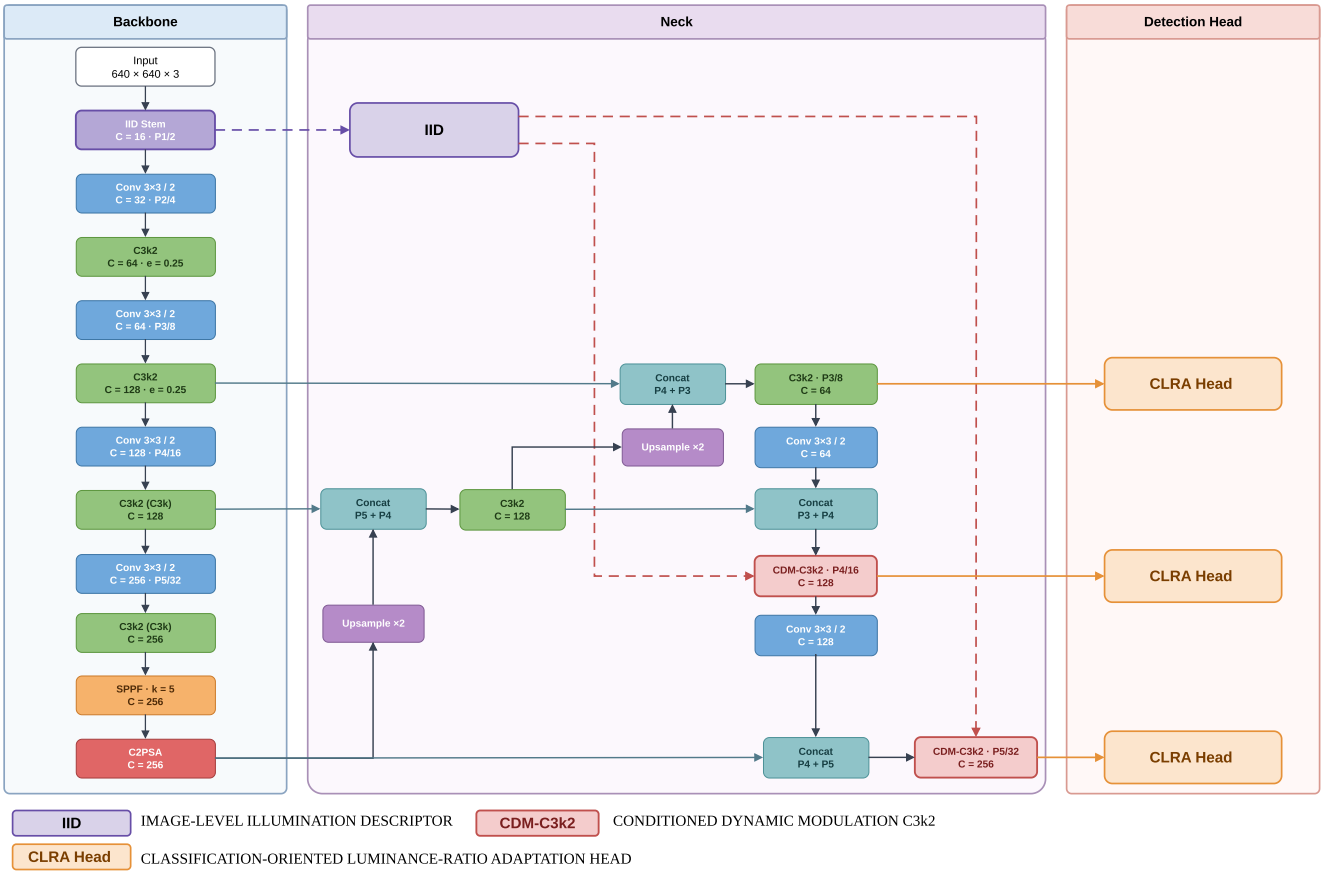


FIGURE 2. **Architecture of ICM-YOLO.**

for evaluating illumination-aware conditioning with fewer architectural confounds.

B. ICM-YOLO FRAMEWORK

ICM-YOLO introduces illumination awareness at selected points of the YOLO26n pipeline while preserving its backbone, bidirectional feature aggregation, and three-scale prediction structure. This design maintains a controlled correspondence with the baseline and avoids the cost of an independent image-enhancement network.

As shown in Figure 2, IID is computed at the network entrance and shared with selected neck stages. CDM-C3k2 replaces the final C3k2 blocks at P4 and P5, where the descriptor conditions semantically enriched features. P3 retains the baseline neck block to preserve its high-resolution detail. CLRA Heads operate at P3, P4, and P5 and adapt the classification features immediately before prediction, while the regression branches receive the original pyramid features.

The resulting information flow separates three roles: IID represents the global illumination state, CDM-C3k2 adapts deeper neck features, and the CLRA Head handles spatial exposure variation for category prediction. The dashed connections in Figure 2 denote conditioning signals rather than an additional detection stream.

C. IMAGE-LEVEL ILLUMINATION DESCRIPTOR

IID obtains an image-level conditioning signal without interrupting the baseline feature stream. For an input batch $\mathbf{I}$, the primary branch performs the original stem transformation and returns the first backbone feature $\mathbf{F}_1$. In parallel, an auxiliary branch characterizes illumination from two complementary sources. A learnable encoder captures global appearance cues, whereas a statistical encoder summarizes the brightness distribution using the mean, dispersion, and two generic quantiles. Both representations are projected into a common latent space and combined to form the descriptor $\mathbf{d}$.

The complete procedure is summarized in Algorithm 1. Here, $\mathcal{P}$ denotes adaptive pooling, $\mathcal{G}$ denotes global average pooling, $\mathcal{E}_\theta$ is the learnable descriptor encoder, and ρ_c is the grayscale coefficient associated with channel c. Because the input image is normalized to $[0, 1]$, the luminance map $\mathbf{Y} \in [0, 1]^{H\times W}$. The quantile levels q_l and q_h satisfy $0 < q_\mathrm{l} < q_\mathrm{h} < 1$. This notation keeps the description independent of a particular input resolution, descriptor width, or pooling size.

Algorithm 1 IID Forward Procedure

Input: image batch $\mathbf{I}$; stem operator $\mathcal{S}$; descriptor encoder $\mathcal{E}_\theta$; statistical projection $(\mathbf{W}_s, \mathbf{b}_s)$.
Output: backbone feature $\mathbf{F}_1$ and descriptor $\mathbf{d}$.
$\mathbf{F}_1 \leftarrow \mathcal{S}(\mathbf{I})$
$\mathbf{Y} \leftarrow \sum_c \rho_c \mathbf{I}_c$
$\mathbf{s} \leftarrow [\mathrm{Mean}(\mathbf{Y}), \mathrm{Std}(\mathbf{Y}), Q_{q_l}(\mathbf{Y}), Q_{q_h}(\mathbf{Y})]^\mathsf{T}$
$\mathbf{Z} \leftarrow \mathcal{P}(\mathbf{I})$
$\mathbf{z} \leftarrow \mathcal{G}(\mathcal{E}_\theta(\mathbf{Z}))$
$\mathbf{d} \leftarrow \mathbf{z} + \mathbf{W}_s\mathbf{s} + \mathbf{b}_s$
cache $\mathbf{d}$ for conditioned neck modules
return $\mathbf{F}_1, \mathbf{d}$

The learned and statistical paths serve different purposes. The learned representation can adapt to illumination patterns not captured by a small set of summary measurements, whereas the statistical vector explicitly describes global brightness and contrast. Their additive combination avoids spatial concatenation with every neck feature. The descriptor is computed once per image and subsequently reused, while the main stem output remains compatible with the pretrained YOLO26n backbone.

D. CONDITIONED DYNAMIC MODULATION

Conditioned dynamic modulation (CDM) transforms a detector feature according to the illumination state of the current input. Let $\mathbf{X}_l \in \mathbb{R}^{B\times C_l\times H_l\times W_l}$ denote the feature at level l, and let $\mathbf{d} \in \mathbb{R}^{B\times D}$ denote the image-level descriptor. A level-specific projection first maps the descriptor into a hidden conditioning representation:

$$\mathbf{u}_l = \phi\left(\mathbf{W}_l^{(1)}\mathbf{d} + \mathbf{b}_l^{(1)}\right), \tag{1}$$

where $\phi(\cdot)$ denotes the SiLU activation. The hidden representation is then converted into a modulation-parameter vector:

$$\mathbf{p}_l = \mathbf{W}_l^{(2)}\mathbf{u}_l + \mathbf{b}_l^{(2)}. \tag{2}$$

The vector $\mathbf{p}_l$ is partitioned into channel-wise scale and shift components, denoted by $\boldsymbol{\gamma}_l$ and $\boldsymbol{\beta}_l$. After broadcasting them over the spatial dimensions, CDM produces

$$\widehat{\mathbf{X}}_l = \mathbf{X}_l \odot \left(\mathbf{1} + \boldsymbol{\gamma}_l\right) + \boldsymbol{\beta}_l. \tag{3}$$

Equation (3) instantiates the residual FiLM formulation. Figure 3 illustrates descriptor projection, parameter partition, spatial broadcasting, and residual affine transformation. The identity term in (3) expresses the learned scale as a deviation from the unconditioned feature rather than as an unrestricted replacement. The output projection is initialized to produce neutral modulation, so the initial forward behavior is equivalent to the corresponding baseline transformation. During optimization, each selected level learns an independent mapping from the shared descriptor to the channel responses required by that semantic level.

Unlike spatial attention, CDM does not predict a dense mask for every feature location. Its conditioning coefficients

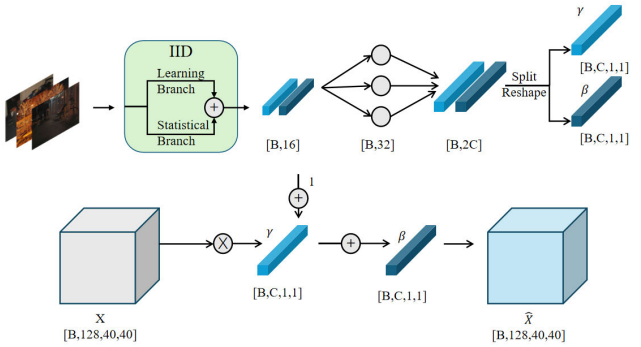


FIGURE 3. Structure of conditioned dynamic modulation.

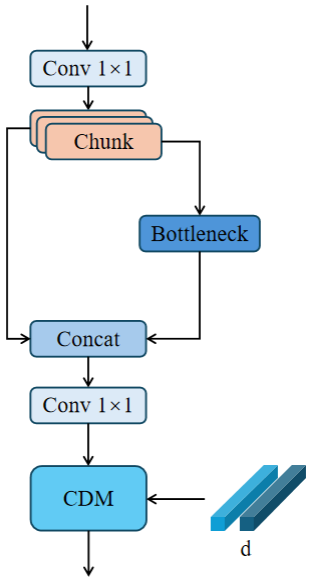


FIGURE 4. Structure of the CDM-C3k2 module.

are shared spatially within a channel and vary across input images. This formulation represents global changes in exposure and contrast while keeping the spatial organization of the detector feature unchanged. It also avoids repeatedly estimating the same image-level state at different neck stages.

E. CDM-C3k2

CDM-C3k2 embeds the conditioning operation into a standard C3k2 block while retaining its input–output interface. Denote the original C3k2 transformation at level l by $\mathcal{B}_l$. Its intermediate output is

$$\mathbf{Z}_l = \mathcal{B}_l(\mathbf{X}_l)\,. \tag{4}$$

The output of CDM-C3k2 is subsequently defined as

$$\mathbf{Y}_l = \mathcal{M}_l(\mathbf{Z}_l, \mathbf{d})\,, \tag{5}$$

where $\mathcal{M}_l$ denotes the CDM operator defined above. This serial arrangement allows C3k2 to complete its local feature transformation and cross-stage aggregation before image-dependent channel adaptation is applied.

As shown in Figure 4, the unchanged C3k2 path is separated from the descriptor-conditioned branch. CDM-C3k2 is

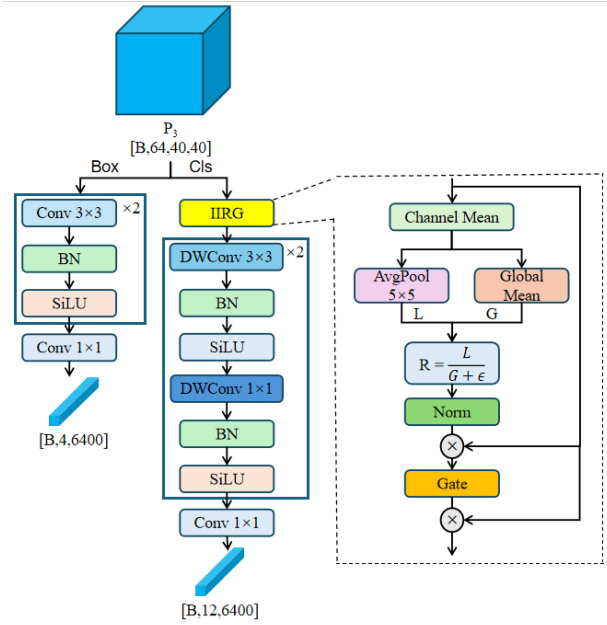


FIGURE 5. Structure of the CLRA Head.

inserted at the final P4 and P5 outputs of the bidirectional neck. These levels contain semantically enriched features for the medium- and large-scale prediction branches. P3 retains the baseline C3k2 block because its high-resolution representation carries fine boundaries and textures that can be sensitive to illumination-dependent amplification.

The module preserves the tensor shape of its baseline counterpart. It can therefore replace selected C3k2 blocks without altering subsequent downsampling, concatenation, or detection interfaces. The baseline path also remains recoverable through neutral modulation, which facilitates initialization from pretrained YOLO26n weights and reduces abrupt changes to the feature distribution at the beginning of training.

F. CLASSIFICATION-ORIENTED LUMINANCE-RATIO ADAPTATION HEAD

The CLRA Head complements IID-based global conditioning with spatially varying adaptation at the prediction stage. Low-light scenes frequently contain dark backgrounds, locally illuminated objects, and saturated light sources within the same image. The module therefore derives an activation-based luminance proxy independently at each pyramid level and uses it to recalibrate category evidence. As shown in Figure 5, the classification pathway performs luminance-proxy ratio estimation, gate prediction, residual recalibration, and path blending, whereas the regression pathway bypasses these photometric operations.

For a feature $\mathbf{X}_l$, an activation-based luminance proxy is obtained by channel averaging:

$$\mathbf{L}_l = \frac{1}{C_l}\sum_{c=1}^{C_l}\mathbf{X}_{l,c}. \tag{6}$$

Let $\mathcal{A}_k$ denote local average pooling with a generic neighborhood size k, and let $\mathcal{G}$ denote global average pooling. The local-to-global ratio of this luminance proxy is

$$\mathbf{R}_l = \frac{\mathcal{A}_k(\mathbf{L}_l)}{\mathcal{G}(\mathbf{L}_l)+\epsilon}. \tag{7}$$

The ratio is normalized spatially to maintain a comparable average response across images:

$$\widehat{\mathbf{R}}_l = \frac{\mathbf{R}_l}{\mathcal{G}(\mathbf{R}_l)+\epsilon}, \tag{8}$$

where ϵ is a small positive constant that prevents division by zero. The ratio path is detached during backpropagation. It serves as a conditioning cue and does not encourage the detector to manipulate the activation-based proxy merely to change the gate. The normalized ratio is broadcast along the channel dimension and combined with the feature before a lightweight mapping $\mathcal{H}_l$ predicts the gate:

$$\mathbf{G}_l = \sigma\left(\mathcal{H}_l\left(\mathbf{X}_l \odot \widehat{\mathbf{R}}_l\right)\right). \tag{9}$$

Here, σ is the sigmoid function. A learnable scale α_l controls the amplitude of the residual recalibration:

$$\widetilde{\mathbf{X}}_l = \mathbf{X}_l \odot (\mathbf{1} + \alpha_l \mathbf{G}_l). \tag{10}$$

For levels that use residual path blending, the classification feature is

$$\mathbf{X}_l^{\text{cls}} = \mathbf{X}_l + \omega_l\left(\widetilde{\mathbf{X}}_l - \mathbf{X}_l\right). \tag{11}$$

The prediction configuration determines whether a level uses the directly recalibrated feature or the blended form. This distinction permits conservative adaptation at detail-sensitive scales while allowing stronger learned integration at semantically deeper scales. The scale coefficients and path weights are implementation parameters rather than fixed components of the mathematical definition.

Illumination adaptation is intentionally excluded from the localization stream. The regression feature is therefore defined as

$$\mathbf{X}_l^{\text{box}} = \mathbf{X}_l. \tag{12}$$

The classification and regression operators consume $\mathbf{X}_l^{\text{cls}}$ and $\mathbf{X}_l^{\text{box}}$, respectively. This asymmetric design reflects the different sensitivities of the two tasks. Category recognition depends strongly on texture, color, and contrast, all of which vary with exposure. Bounding-box regression relies more directly on spatial geometry and boundary placement, for which unnecessary photometric recalibration may be disruptive. By retaining the original regression feature, the CLRA Head introduces illumination awareness without coupling localization to the estimated luminance-proxy ratio.

III. EXPERIMENTS

A. DATASETS

1) ExDark

ExDark contains 7,363 low-light images from 12 object categories [27]. This study adopts the official training and

TABLE 1. Dataset statistics and splits.

Dataset	Modality	Train	Val.	Classes
ExDark	Visible	3,000	1,800	12
M3FD	Visible only	2,100	2,100	6

validation lists (3,000/1,800 images), whereas the remaining 2,563 images form a third list that is not used in this study. One validation image is excluded because its annotation contains an out-of-range normalized box width (1.075), leaving 1,799 images and 6,211 instances. The evaluated per-class counts are: Bicycle 239; Boat 344; Bottle 506; Bus 209; Car 770; Cat 173; Chair 630; Cup 436; Dog 192; Motorbike 340; People 1,954; and Table 418. All detectors evaluated in this article use the same split and filtering rule, so the reported comparisons are internally controlled. Because published studies adopt different ExDark partitions, the absolute values obtained under this protocol are not directly comparable with previously reported results.

2) M3FD

M3FD provides spatially aligned visible and infrared images acquired across diverse scenes, illumination conditions, seasons, and weather conditions [28]. The present study uses the visible modality only; consequently, the reported experiment provides an additional evaluation on visible-light data rather than an assessment of multispectral fusion. The adopted split contains 2,100 training images and 2,100 validation images from six categories: People, Car, Bus, Truck, Motorcycle, and Lamp.

B. EXPERIMENTAL ENVIRONMENT

All experiments are implemented in PyTorch with the Ultralytics detection framework and conducted on an NVIDIA L20 GPU with 46 GB of memory. The models use 640×640 inputs, and both YOLO26n and ICM-YOLO are initialized from the same official COCO-pretrained checkpoint, `yolo26n.pt` [12], [29]. The IID quantile levels are set to $q_{\mathrm{l}} = 0.25$ and $q_{\mathrm{h}} = 0.75$.

C. EVALUATION METRICS

1) EVALUATION METRICS

Detection performance is evaluated using precision, recall, average precision (AP), mAP@50, and mAP@50:95. Let TP, FP, and FN denote the numbers of true-positive, false-positive, and false-negative predictions under a specified confidence and intersection-over-union (IoU) criterion. Precision is defined as

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \tag{13}$$

whereas recall is defined as

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{14}$$

Precision measures the reliability of positive detections, while recall measures the proportion of annotated objects recovered by the detector.

For object category c and IoU threshold τ, AP is the area under the interpolated precision–recall curve:

$$\mathrm{AP}_c(\tau) = \int_0^1 p_c(r;\tau)\,\mathrm{d}r, \tag{15}$$

where $p_c(r;\tau)$ denotes interpolated precision as a function of recall r. The mean AP at threshold τ is

$$\mathrm{mAP}(\tau) = \frac{1}{N_c}\sum_{c=1}^{N_c}\mathrm{AP}_c(\tau), \tag{16}$$

where N_c is the number of object categories. Accordingly, mAP@50 evaluates detections at $\tau = 0.50$. The stricter mAP@50:95 averages $\mathrm{mAP}(\tau)$ over the standard threshold set from 0.50 to 0.95 with an interval of 0.05:

$$\mathrm{mAP}_{50:95} = \frac{1}{|\mathcal{T}|}\sum_{\tau\in\mathcal{T}}\mathrm{mAP}(\tau), \tag{17}$$

where $\mathcal{T} = \{0.50, 0.55, \ldots, 0.95\}$. The latter metric jointly reflects classification accuracy and localization quality across progressively stricter overlap requirements. In the text and tables, mAP(0.50) and $\mathrm{mAP}_{50:95}$ are reported as mAP@50 and mAP@50:95, and their class-wise counterparts as AP_{50} and $\mathrm{AP}_{50:95}$.

D. COMPARISON WITH YOLO DETECTORS ON ExDark

Table 2 compares nano- or tiny-scale detectors on the same ExDark validation split. YOLO26n serves as the controlled host baseline for evaluating IID, CDM-C3k2, and the CLRA Head, whereas YOLOv8n, YOLOv9t, YOLOv10n, YOLO11n, and YOLO12n serve as established, publicly available reference detectors.

As shown in Table 2, ICM-YOLO attains the highest overall accuracy, with 63.75% mAP@50 and 38.94% mAP@50:95. The corresponding YOLO26n results are 62.43% and 38.59%, while recall improves from 56.65% for YOLO26n to 59.25% for ICM-YOLO. YOLOv9t provides the closest comparison among the remaining detectors, reaching 63.21% mAP@50 and 38.87% mAP@50:95. The larger gain at IoU 0.50 is consistent with improved detection at a lenient overlap threshold, although the aggregate metrics do not isolate classification and localization effects.

YOLOv8n, YOLOv10n, YOLO11n, and YOLO12n obtain mAP@50 values between 56.70% and 58.36%, while YOLOv9t and YOLO26n provide closer reference results. This distribution indicates that performance on ExDark does not follow model generation alone. Among the reference detectors, YOLOv9t has the fewest parameters, whereas YOLO26n has the lowest operation count. In contrast, ICM-YOLO provides the best mAP@50, mAP@50:95, and recall, demonstrating a favorable balance between detection accuracy and model complexity.

TABLE 2. Detection performance on ExDark.

Model	Pretrained Weights	mAP@50	mAP@50:95	Recall	Params (M)	GFLOPs
YOLOv8n [7]	YOLOv8n	58.36	34.59	52.41	3.01	8.21
YOLOv9t [8]	YOLOv9t	63.21	38.87	58.65	**2.01**	7.86
YOLOv10n [9]	YOLOv10n	57.10	33.59	50.63	2.71	8.42
YOLO11n [10]	YOLO11n	57.53	33.80	52.48	2.59	6.45
YOLO12n [11]	YOLO12n	56.70	33.48	52.29	2.57	6.49
YOLO26n [12]	YOLO26n	62.43	38.59	56.65	2.51	**5.80**
ICM-YOLO	YOLO26n	**63.75**	**38.94**	**59.25**	2.56	5.89

Model names follow the official releases. The suffixes n and t denote the nano and tiny scales, respectively, and YOLO12n denotes the nano YOLOv12 model [11] as distributed in the Ultralytics framework. The Pretrained Weights column lists the official COCO-pretrained checkpoint used for initialization; for example, YOLO26n denotes `yolo26n.pt`.

E. ABLATION STUDIES ON ExDark

Table 3 examines the effect of the CLRA Head and the placement of CDM-C3k2. Adding the CLRA Head to YOLO26n raises mAP@50 from 62.43% to 63.04% and mAP@50:95 from 38.59% to 38.90%; recall also rises from 56.65% to 58.09%. The CDM-C3k2 results further show that placement is important. Applying the module to all three pyramid levels produces lower accuracy than retaining the original P3 block and conditioning P4/P5 only. The latter configuration achieves the best results, reaching 63.75% mAP@50, 38.94% mAP@50:95, and 59.25% recall.

The CLRA Head acts only on the classification pathway and leaves box regression unchanged. Its improvement over YOLO26n suggests that the activation-based luminance proxy can provide useful category evidence without directly modifying localization features. The accompanying increase in recall is also consistent with more low-confidence objects being retained by the classifier.

The two CDM-C3k2 placements exhibit notably different behavior. When the module is extended to P3, mAP@50 falls to 61.93% and mAP@50:95 to 37.19%; restricting it to P4/P5 raises these values to 63.75% and 38.94%. Since P3 carries higher-resolution information used for small objects and fine boundaries, preserving its baseline block appears beneficial. Within the tested configurations, restricting illumination conditioning to the deeper semantic features at P4 and P5 performs better.

Taken together, the ablation results show that adding CDM-C3k2 at P4/P5 to the CLRA-equipped detector produces the best tested configuration. The CLRA Head adjusts spatially varying classification responses close to prediction, whereas CDM-C3k2 uses IID to condition deeper neck features. Their combined configuration produces the highest recall and aggregate AP in Table 3, supporting classification-oriented adaptation and selective neck conditioning.

F. CLASS-WISE PERFORMANCE ON ExDark

Tables 4 and 5 report category-level AP for the seven detectors evaluated in Table 2. The 12 categories are divided equally between the two tables, with adjacent AP@50 and AP@50:95 columns for each category. All category-level values are obtained from the same independent evaluations as the aggregate metrics.

Across the evaluated detectors, Bus and Bicycle generally have higher AP@50 values, whereas Table, Bottle, and Chair have lower values. YOLOv9t records the highest results for Car and People, while YOLO26n records the highest result for Bus. The category-level variation indicates that similar aggregate mAP values can correspond to different class-specific behavior.

ICM-YOLO records the highest AP@50 for Bicycle, Boat, Cat, Cup, Dog, Motorbike, and Table, and also exceeds YOLO26n for People. The most visible gains occur for Cup, whose AP@50 rises from 56.91% to 62.54%, Motorbike, from 67.09% to 71.31%, and Bicycle, from 75.07% to 78.88%. The trend is not uniform across categories: Chair decreases from 55.58% to 49.47%, and Bus from 81.74% to 79.20%. A similar class-dependent pattern is observed at AP@50:95.

Chair exhibits the largest AP@50 reduction relative to YOLO26n. Table 6 traces this category through the ablation configurations of Table 3. Adding the CLRA Head alone leaves Chair AP@50 essentially unchanged (55.73% versus 55.58%), whereas both CDM-C3k2 placements reduce it to approximately 50% (49.94% at P3/P4/P5 and 49.47% at P4/P5). The reduction therefore coincides with image-level neck conditioning rather than with prediction-stage recalibration. It also arises mainly from recall: Chair recall decreases from 50.22% to 38.66%, whereas precision increases from 65.45% to 74.58%. ICM-YOLO thus responds more conservatively to chairs, leaving more true instances below the confidence threshold rather than producing additional false positives.

G. COMPLEXITY AND INFERENCE EFFICIENCY

Relative to YOLO26n, ICM-YOLO adds only 0.05 M parameters and 0.09 GFLOPs, corresponding to increases of approximately 2.0% and 1.6%, respectively. Table 7 reports the measured latency and throughput on an NVIDIA L20 before and after convolution–normalization fusion.

Among the compared models, YOLO26n has the lowest operation count, while YOLOv9t has the fewest parameters. Parameter count and GFLOPs do not determine runtime uniquely because depthwise operations, branching, memory access, and framework-level kernels have different execution characteristics. Accordingly, the 5.89 GFLOPs of

TABLE 3. Ablation results on ExDark.

Model	Pretrained Weights	mAP@50	mAP@50:95	Recall	Params (M)	GFLOPs
YOLO26n [12]	YOLO26n	62.43	38.59	56.65	2.51	5.80
YOLO26n+CLRA Head	YOLO26n	63.04	38.90	58.09	2.53	5.87
YOLO26n+CLRA Head+CDM-C3k2@P3/P4/P5	YOLO26n	61.93	37.19	56.78	2.56	5.89
ICM-YOLO (CDM-C3k2@P4/P5)	YOLO26n	**63.75**	**38.94**	**59.25**	2.56	5.89

TABLE 4. Class-wise AP (%) on ExDark: bicycle–cat.

Model	Bic.		Boat		Bot.		Bus		Car		Cat	
	AP_{50}	$AP_{50:95}$	AP_{50}	$AP_{50:95}$	AP_{50}	$AP_{50:95}$	AP_{50}	$AP_{50:95}$	AP_{50}	$AP_{50:95}$	AP_{50}	$AP_{50:95}$
YOLOv8n [7]	72.62	45.56	53.41	25.48	49.27	28.15	74.62	56.40	70.71	43.52	50.09	28.80
YOLOv9t [8]	72.14	42.70	54.10	26.38	**53.17**	**31.38**	78.86	61.62	**76.05**	**49.14**	60.15	37.04
YOLOv10n [9]	71.94	41.45	49.17	23.59	47.63	28.82	77.43	58.31	68.01	42.37	48.20	27.59
YOLO11n [10]	64.94	37.21	50.85	23.56	52.01	28.93	77.31	58.27	70.36	43.95	47.34	23.77
YOLO12n [11]	71.39	41.18	49.84	22.70	44.77	25.76	77.15	58.63	67.30	42.27	42.34	25.57
YOLO26n [12]	75.07	46.43	55.97	26.57	49.23	28.66	**81.74**	**64.51**	71.88	46.55	59.59	36.89
ICM-YOLO	**78.88**	**49.15**	**57.71**	**28.21**	49.13	28.07	79.20	61.60	72.29	46.01	**62.42**	**37.58**

For each category, the left column reports AP_{50}, that is, AP at an IoU threshold of 0.50, and the right column reports $AP_{50:95}$, that is, AP averaged over IoU thresholds from 0.50 to 0.95. Bic.: Bicycle; Bot.: Bottle.

TABLE 5. Class-wise AP (%) on ExDark: chair–table.

Model	Chair		Cup		Dog		M.bike		People		Table	
	AP_{50}	$AP_{50:95}$	AP_{50}	$AP_{50:95}$	AP_{50}	$AP_{50:95}$	AP_{50}	$AP_{50:95}$	AP_{50}	$AP_{50:95}$	AP_{50}	$AP_{50:95}$
YOLOv8n [7]	49.25	27.71	54.16	32.42	56.91	35.98	63.28	35.32	69.18	35.34	36.83	20.37
YOLOv9t [8]	55.27	31.63	59.83	37.98	61.28	39.36	66.41	38.81	**72.10**	**39.62**	49.12	**30.80**
YOLOv10n [9]	48.08	26.08	53.38	32.80	55.19	36.08	60.58	31.36	63.92	33.17	41.67	21.45
YOLO11n [10]	44.94	25.20	56.70	35.50	58.33	36.99	58.90	33.86	69.17	36.26	39.46	22.14
YOLO12n [11]	45.37	24.61	51.14	31.69	53.80	33.96	64.24	34.94	66.38	34.15	46.63	26.27
YOLO26n [12]	**55.58**	**32.62**	56.91	36.74	59.31	40.74	67.09	37.55	68.97	37.85	47.86	28.00
ICM-YOLO	49.47	28.51	**62.54**	**39.72**	**61.30**	**41.72**	**71.31**	**40.31**	70.88	37.46	**49.92**	28.95

For each category, the left column reports AP_{50}, that is, AP at an IoU threshold of 0.50, and the right column reports $AP_{50:95}$, that is, AP averaged over IoU thresholds from 0.50 to 0.95. M.bike: Motorbike.

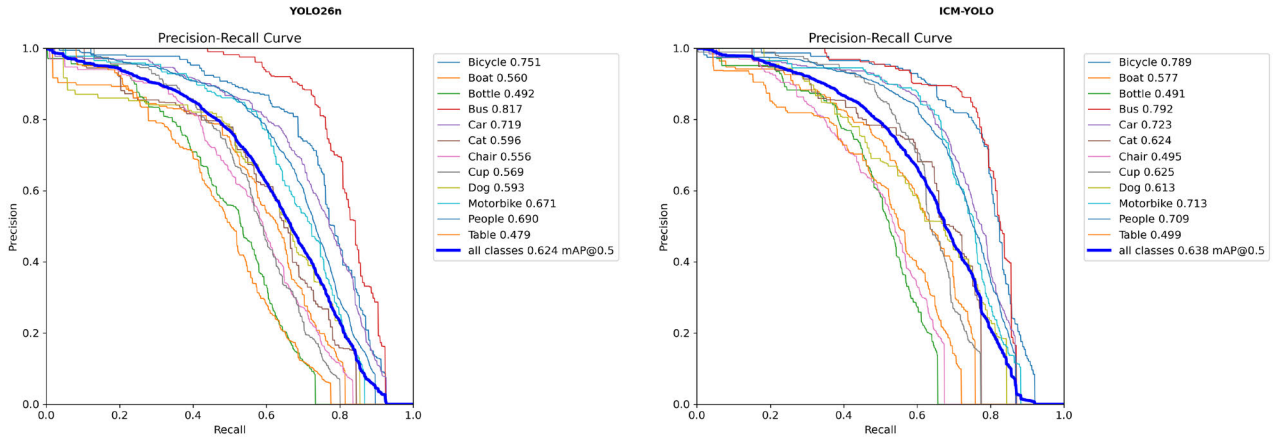


FIGURE 6. Precision–recall curves on ExDark.

ICM-YOLO quantify its arithmetic cost but do not fully represent the runtime associated with descriptor statistics and feature-wise modulation.

As shown in Table 7, ICM-YOLO reaches 77.31 FPS before fusion and 128.94 FPS after fusion on the NVIDIA L20. Although illumination conditioning introduces additional latency, the detector maintains high throughput while delivering the best aggregate accuracy on ExDark.

H. QUALITATIVE RESULTS ON ExDark

Fig. 6 presents the class-wise precision–recall curves, and Fig. 7 shows four selected ExDark validation images with



FIGURE 7. Qualitative comparisons on selected ExDark images with positive detection changes.



FIGURE 8. Qualitative comparisons on selected ExDark images with negative detection changes.

pronounced positive detection changes under class-aware matching.

TABLE 6. Chair results across ablation configurations on ExDark.

Model	AP_{50}	$AP_{50:95}$	Prec.	Rec.
YOLO26n	55.58	32.62	65.45	50.22
+CLRA Head	**55.73**	**32.87**	59.80	**54.60**
+CLRA Head+CDM-C3k2@P3/P4/P5	49.94	29.47	72.78	42.38
ICM-YOLO (CDM-C3k2@P4/P5)	49.47	28.51	**74.58**	38.66

Precision (Prec.) and recall (Rec.) are measured at the confidence threshold that maximizes the mean F1 score over all categories.

TABLE 7. Inference efficiency on an NVIDIA L20.

Model	Unfused		Fused	
	Lat. (ms)	FPS	Lat. (ms)	FPS
YOLO26n [12]	**9.26**	**108.02**	**5.87**	**170.41**
ICM-YOLO	12.94	77.31	7.76	128.94

The precision–recall curves characterize performance over a range of confidence thresholds rather than at one operating point. For several categories, the curve of ICM-YOLO retains higher precision over part of the recall range, consistent with the aggregate recall difference in Table 2. The lower Chair curve over a broader interval indicates that the category-level reduction is not restricted to one confidence threshold.

Across the four images in Fig. 7, YOLO26n yields 29 matched true positives, whereas ICM-YOLO yields 50. The per-image changes are 1 → 7, 9 → 15, 12 → 17, and 7 → 11. The scenes include a dark road, a crowded event, a rainy street, and a bicycle-dense night scene. In these examples, additional People, Bicycle, Motorbike, and vehicle detections exceed the confidence threshold with IID, CDM-C3k2, and the CLRA Head. This observation is consistent with, but does not establish, the proposed complementary roles of image-level neck conditioning and prediction-stage recalibration. These four images are provided for illustration; generalization is assessed from the complete validation and class-wise results, the ablation placements, the failure cases in Fig. 8, and the visible-only M3FD evaluation.

I. EXTENDED EVALUATION ON M3FD

Table 8 compares YOLO26n and ICM-YOLO on the visible-only M3FD validation split.

The M3FD results remain close at IoU 0.50. ICM-YOLO reaches 76.16% mAP@50, compared with 76.03% for YOLO26n, and their recall values are 69.52% and 69.25%, respectively. At the stricter metric, ICM-YOLO reaches 49.97% mAP@50:95, compared with 49.61% for YOLO26n. These visible-only results therefore constitute a small gain over YOLO26n.

The conditioning modules increase the parameter count from 2.51 M to 2.55 M and the operation count from 5.78 to 5.87 GFLOPs. These values are slightly lower than their ExDark counterparts in Table 2 because the M3FD head predicts 6 classes instead of 12; the relative

TABLE 8. Detection performance on visible M3FD.

Model	Pretrained Weights	mAP@50	mAP@50:95	Recall	Params (M)	GFLOPs
YOLO26n [12]	YOLO26n	76.03	49.61	69.25	**2.51**	**5.78**
ICM-YOLO	YOLO26n	**76.16**	**49.97**	**69.52**	2.55	5.87

TABLE 9. IID statistics on failure, success, and ExDark val images.

Group	Mean	Std	$Q_{0.25}$	$Q_{0.75}$	$\mathbf{Y}>0.9$ (%)	$\|\mathbf{d}\|_2$
Failure (4)	0.196	0.186	0.066	0.252	1.30	0.331
Success (4)	0.173	0.167	0.056	0.241	1.17	0.321
ExDark val	0.162	0.168	0.049	0.218	1.31	0.322

overhead of the proposed modules is unaffected. With this limited overhead, ICM-YOLO yields a small gain on visible-only M3FD.

IV. DISCUSSION

A. ANALYSIS OF FAILURE CASES

Fig. 8 shows four ExDark images with negative detection changes. Across these images, YOLO26n produces 44 matched true positives and 9 false positives, whereas ICM-YOLO produces 27 matched true positives and 22 false positives. These scenes contain strong local light sources, severe occlusion, repeated structures, and cluttered backgrounds.

Table 9 reports the four luminance statistics of $\mathbf{s}$, the proportion of pixels whose normalized luminance $\mathbf{Y}$ exceeds 0.9, and $\|\mathbf{d}\|_2$ for these failure images, the success images of Fig. 7, and the complete validation set, all measured with the trained ICM-YOLO model. The failure group is brighter than the validation average in the upper part of its luminance range, with a mean luminance of 0.196 and a third quartile of 0.252, exceeding the corresponding validation values by 21% and 16%. Its mean highlight ratio, in contrast, remains comparable to the validation average, and its descriptor norm increases by less than 3%. Because each group contains only four images, these differences are descriptive rather than statistically conclusive. They nevertheless suggest that severe local saturation perturbs the global illumination condition in individual scenes, although it does not account for all observed failures. Occlusion, small objects, and background clutter also contribute, which is consistent with the reduced AP@50 obtained for Chair and Bus. More spatially robust illumination descriptors and category-aware calibration therefore represent promising directions for future work.

B. MULTI-SEED EVALUATION

Tables 2 and 8 report a single run of each configuration. To quantify how much of the reported difference can be attributed to initialization variance, YOLO26n and ICM-YOLO were retrained as paired runs on three prespecified seeds, $\{0, 1, 2\}$, under identical data splits and pretrained checkpoints and a common 200-epoch schedule without early stopping. This reduced schedule is applied identically to both models, so the paired differences remain controlled although the absolute values are not directly comparable with the main tables. Table 10 reports the independent `best.pt` evaluation of each run together with the mean $\pm$ sample standard deviation over the three seeds, and no seed was excluded. On ExDark, ICM-YOLO attains $64.42 \pm 0.71\%$ mAP@50 and YOLO26n $63.70 \pm 0.44\%$, a mean difference of 0.72 percentage points, with ICM-YOLO higher for two of the three seeds. On visible-only M3FD the corresponding values are $76.25 \pm 0.37\%$ and $76.00 \pm 0.28\%$, while mAP@50:95 and recall differ by 0.01 and 0.10 percentage points.

TABLE 10. Three-seed evaluation of YOLO26n and ICM-YOLO.

Model	Metric	Seed 0	Seed 1	Seed 2	Mean$\pm$Std
ExDark					
YOLO26n	mAP@50	63.72	63.26	64.14	63.70 ± 0.44
	mAP@50:95	39.45	39.27	39.87	39.53 ± 0.30
	Recall	59.09	57.85	58.23	58.39 ± 0.64
ICM-YOLO	mAP@50	64.65	65.00	63.62	64.42 ± 0.71
	mAP@50:95	40.05	40.50	39.47	40.01 ± 0.52
	Recall	59.57	59.79	58.73	59.37 ± 0.56
Visible M3FD					
YOLO26n	mAP@50	75.98	75.73	76.29	76.00 ± 0.28
	mAP@50:95	49.53	49.06	49.51	49.37 ± 0.26
	Recall	69.18	67.69	68.45	68.44 ± 0.74
ICM-YOLO	mAP@50	76.15	76.67	75.95	76.25 ± 0.37
	mAP@50:95	49.08	49.82	49.18	49.36 ± 0.40
	Recall	69.01	68.38	67.65	68.34 ± 0.68

C. EFFICIENCY AND DEPLOYMENT CONSIDERATIONS

ICM-YOLO adds 0.05 M parameters and 0.09 GFLOPs over YOLO26n, corresponding to increases of approximately 2.0% and 1.6%, respectively. As reported in Table 7, it achieves 77.31 FPS before fusion and 128.94 FPS after fusion on the NVIDIA L20.

The small increase in theoretical complexity shows that IID, CDM-C3k2, and the CLRA Head remain lightweight. The larger runtime difference relative to the GFLOP increase is mainly associated with descriptor statistics, dynamic modulation, and additional feature operations. Further kernel fusion and deployment-oriented optimization could reduce this implementation overhead.

V. CONCLUSION

This article presented ICM-YOLO, a compact low-light detector integrating IID, CDM-C3k2, and the CLRA Head for complementary global and spatial illumination adaptation. On ExDark, ICM-YOLO achieves 63.75% mAP@50 and 38.94% mAP@50:95, compared with 62.43% and 38.59% for YOLO26n, and records higher AP@50 in nine of the twelve categories. The added modules increase parameters and operations by approximately 2.0% and 1.6%, respectively, while achieving 77.31 FPS on the NVIDIA L20. On visible-only M3FD, ICM-YOLO yields a small gain over YOLO26n. The gains are not uniform across categories: Chair, which frequently appears in cluttered indoor scenes with bright backgrounds, decreases under image-level conditioning, motivating spatially localized or category-aware conditioning in future work. These results show that selective illumination conditioning improves the reported low-light detection accuracy with limited computational overhead.

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